NOVELTY-SORTED RADAR RESEARCH PORTFOLIO

Repository-wise ranking and individual technical explanations

25 concepts | 4 repositories | sorted by novelty score

Each concept progresses from intuition to equations, implementation, numerical evidence, and advanced theoretical foundation.

How the portfolio is organized

Each repository has its own independent ranking table. Concepts are sorted from highest to lowest novelty within that repository. Detailed concept sheets follow the same repository-wise order. Within each sheet, the explanation progresses from foundational intuition to advanced theory.

READING ORDER: FOUNDATIONAL TO ADVANCED

1. Intuition (13-year-old level)
2. Baseline formulation
3. Novel formulation
4. Technical mechanism and implementation
5. Concrete numerical example
6. Theoretical foundation (PhD level)

Novelty scores are the supplied scores. They represent comparative research novelty, not independently audited patentability, publication priority, or production readiness.

Repository 1 Novelty Ranking

Central Object Tracker

`CORECOMP/ALSW/OT_ObjectTracking`

Rank	Research breakthrough	Score	Target file
1	**Virtual Aperture Specular Tracking**	80.0	`Msmt_Update_ObjTrks_CCA_Moveable.cpp`
2	**Rank-1 Specular Invariant and Bearing Correction**	74.0	`f360_detect_multipath.cpp`
3	**Kinematics-Aware Adaptive Gating**	71.0	`f360_determine_association_hypothesis.cpp`
4	**Tangent-Mirror Hermite Linearization**	68.0	`f360_is_reflective_guardrail_track.cpp`
5	**Distribution-Free Conformal Gating**	67.0	`f360_determine_association_hypothesis.cpp`
6	**Spectral Entropy Covariance Scaling**	67.0	`Msmt_Update_ObjTrks_CTCA.cpp`
7	**Contamination-Budgeted Frozen Rank Gate**	63.0	`f360_determine_association_hypothesis.cpp`
8	**Doppler-Invariant Ghost Disambiguation**	63.0	`f360_detect_multipath.cpp`
9	**IMM Acceleration Gating (Refuted Baseline)**	55.0	`f360_determine_association_hypothesis.cpp`

Repository 2 Novelty Ranking

Satellite Radar DSP Firmware

`Core_Radar_Gen8_iND13400`

Rank	Research breakthrough	Score	Target file
1	**Dual-Horizon Fast Dynamic Alignment**	70.0	`dyn_align_wrapper.c`
2	**Track-Conditioned Residual Beat Estimator**	65.0	`residual_track_ifc.c`
3	**Occupancy-BLUE CFAR with Pfa Control**	62.0	`appl_cfar_ifc.c` / `appl_adaptive_cfar.c`
4	**Range-Adaptive Dual-Loop Clutter CFAR**	62.0	`appl_cfar_ifc.c`
5	**Closed-Form Specular Contact Inversion**	60.0	`ghost_calib.c`
6	**Priority-Ranked CDC Urban Packing**	60.0	`cdc_packing.c / cdc_priority.c`
7	**3-Product Cubic-Exact Residual Stencil**	57.0	`residual_track_ifc.c`

Repository 3 Novelty Ranking

Domain Controller SiL Replay Harness

`Core_RESIM_DC_Emb_Library`

Rank	Research breakthrough	Score	Target file
1	**Preflight DA Header Injection**	70.0	`sil_wrapper.cpp` / `Tracker_VariantA_Wrapper.cpp`
2	**Asynchronous Satellite Yaw Extrapolation**	68.0	`sil_wrapper.cpp`
3	**Yaw-Free Timestamp and Doppler-Bias Separation**	60.0	`sil_wrapper.cpp`
4	**Cross-Satellite Relative Yaw Cancellation**	58.0	`sil_wrapper.cpp`

Repository 4 Novelty Ranking

Distributed Simulation and Cluster Infrastructure

`Core_RESIM_HPCC / Engine`

Rank	Research breakthrough	Score	Target file
1	**4D Doppler-RadarSplat Generative Simulation**	70.0	`models/doppler_radarsplat.py`
2	**Associative Parallel Prefix-Scan Kalman Filter**	64.0	`engine/parallel_kf.py`
3	**Ghost-Conditioned Generative Target Augmentation**	64.0	`models/ghost_augmentation.py`
4	**Nonlinear Parallel Gram-Lifted State Space**	60.0	`engine/gram_lift.py`
5	**Guarded Error-Transfer Replay Certificates**	58.0	`engine/guarded_replay.py`

Virtual Aperture Specular Tracking

NOVELTY SCORE 80.0 / 100 Msmt_Update_ObjTrks_CCA_Moveable.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

A guardrail reflection behaves like a second radar eye. Two viewing directions make sideways position easier to judge, just as two eyes improve depth perception.

2. BASELINE FORMULATION

Single line-of-sight update

```
y_k = z_dir - h(x_k^-)
Multipath is penalized or suppressed through mirror_prob.
```

Tracking RMSE: 1.969 m.

3. NOVEL FORMULATION

Coupled-covariance bistatic update row

```
[y_dir; y_ghost] = [z_dir - h(x_k^-); z_ghost - h_s(x_k^-)]
R_eff = diag(sigma^2) + J_xi Sigma_xi J_xi^T
```

Tracking RMSE: 0.651 m, a 67.0% reduction in error.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Pair validated specular reflections with their parent tracks. Append a synthetic observation row corresponding to a line of sight from the virtual sensor image s*. Propagate shared guardrail uncertainty Sigma_xi through the Jacobian J_xi and use a Joseph-form covariance update. This avoids treating direct and reflected observations as independent and prevents covariance collapse or filter overconfidence.

Implementation checkpoints

- Validate ghost-parent consistency before extending the measurement vector.
- Construct the virtual sensor pose from the estimated reflector plane.
- Augment innovation and measurement Jacobian with the specular row.
- Include reflector-state uncertainty in R_eff.
- Apply numerically stable Joseph covariance update and monitor positive definiteness.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A host radar at (0, 0) tracks a car at (30, 2) m beside a guardrail at y = 3 m. The direct view has weak lateral resolution, approximately sigma_y = 1.8 m. Mirroring the sensor across the rail creates a virtual sensor at (0, 6) m, which observes the opposite flank of the car. The two viewing directions create cross-triangulation and reduce the lateral tracking error from 1.969 m to 0.651 m.

6. THEORETICAL FOUNDATION - PHD LEVEL

Virtual-aperture multistatic estimation; augmented EKF observation manifolds; common reflector uncertainty projected by J_xi Sigma_xi J_xi^T; GDOP reduction with correlated measurements.

Technical caution: Incorrect reflector association can inject a high-leverage synthetic row. Require temporal confirmation, innovation bounds, and covariance floors.

REPOSITORY 1 | RANK 2 | CONCEPT 2/25

Rank-1 Specular Invariant and Bearing Correction

NOVELTY SCORE 74.0 / 100 f360_detect_multipath.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

In a mirror, the apparent object is behind the glass, but the ray reaching the eye comes from a reflected viewing position. The key is to mirror the sensor, not the target.

2. BASELINE FORMULATION

Target-mirror projection

```
p* = p - 2(n_hat^T p - d)n_hat
```

Range error: 0.0 m, but bearing error: 15.19 degrees, equivalent to an 8.0 m lateral bias in the example.

3. NOVEL FORMULATION

Virtual-sensor ray invariant

```
s* = 2d n_hat
p_s = p - 2d n_hat
||p - p_s - 2d n_hat|| <= 1.07e-14
```

Numerical invariant residual is effectively zero.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Correct the arrival direction at the antenna array by expressing the path as a ray from the virtual sensor image to the real target. The baseline mirrors the target, which preserves total path length but reverses the arrival-bearing interpretation. The displacement invariant makes the geometry explicit and improves association between ghost detections and parent tracks.

Implementation checkpoints

- Represent each planar reflector in Hessian normal form.
- Form the virtual sensor image, not a mirrored target, for bearing prediction.
- Use the invariant residual as a unit-test oracle.
- Preserve the correct coordinate-frame and angle-sign conventions.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

For a rail at y = 3 m with d = 3 and n_hat = [0, 1]^T, and a target at (30, 2) m, target mirroring gives (30, 4) m and bearing atan(4/30) = +7.59 degrees. The physical reflected wave is equivalent to a virtual sensor at (0, 6) m and therefore arrives along atan(-4/30) = -7.59 degrees. The baseline sign error is 15.19 degrees.

6. THEORETICAL FOUNDATION - PHD LEVEL

Specular geometry in Hessian normal form; sensor-origin reflection as an SE(2) translation; projective bearing is not preserved by naively mirroring the target endpoint.

Technical caution: Normal-vector sign and sensor-frame conventions can silently flip the result. Add coordinate-frame tests for all vehicle quadrants.

REPOSITORY 1 | RANK 3 | CONCEPT 3/25

Kinematics-Aware Adaptive Gating

NOVELTY SCORE 71.0 / 100 f360_determine_association_hypothesis.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

A catcher uses a small glove area for a straight pitch and a wider area for a sudden curve, but never opens the gate so far that dirt is accepted as the ball.

2. BASELINE FORMULATION

Fixed Mahalanobis threshold

```
d_M^2 = y_k^T S_k^-1 y_k <= gamma_0
gamma_0 = 9.21 (chi-square, 2 DoF, 99% bound)
```

Cut-in recall: 86.3%; maneuver RMSE: 0.283 m.

3. NOVEL FORMULATION

Dynamic kinematic expansion

```
gamma_ad = gamma_0 [1 + alpha ||a_target||/a_max +
eta ||y_k||^2/Tr(S_k)]
Clamp gamma_ad to [9.21, 40.0].
```

Recall: 99.6%; maneuver RMSE: 0.212 m.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Scale the association gate using target lateral acceleration and normalized innovation energy. A hard upper bound prevents uncontrolled spatial expansion in dense clutter, while the acceleration term preserves continuity during genuine cut-in or evasive maneuvers.

Implementation checkpoints

- Estimate acceleration from a smoothed track state, not a single-frame difference.
- Normalize the innovation term to avoid unit dependence.
- Clamp both expansion and contraction.
- Log gate causes so KPI analysis can separate maneuver recovery from clutter admission.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

For lateral acceleration 6.0 m/s^2, a_max = 8.0, alpha = 1.5, eta = 0.2, normalized innovation term 2.1, and d_M^2 = 18.4, the static 9.21 gate rejects the return. The adaptive threshold becomes approximately 23.4 and retains the detection.

6. THEORETICAL FOUNDATION - PHD LEVEL

State-dependent dilation of a Mahalanobis ellipsoid; compensation for truncated process noise during jerk; bounded acceptance regions preserve clutter robustness.

Technical caution: Noisy acceleration estimates may expand gates spuriously. Apply persistence and acceleration-confidence checks.

REPOSITORY 1 | RANK 4 | CONCEPT 4/25

Tangent-Mirror Hermite Linearization

NOVELTY SCORE 68.0 / 100 f360_is_reflective_guardrail_track.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

Two reflections reveal both where a curved mirror is and how it tilts at two points, enough to reconstruct a smooth local curve.

2. BASELINE FORMULATION

Planar reflection or iterative grid search

```
Assume flat barrier: curvature kappa = 0
Alternative: nonlinear Fermat ray-tracing search
```

Curved-rail recall: 19.8%.

3. NOVEL FORMULATION

4 x 4 confluent Vandermonde solve

```
y(x) = a x^3 + b x^2 + c x + d
[y(x1), y'(x1), y(x2), y'(x2)] determines [a,b,c,d]
det(V) = (x2 - x1)^4 != 0
```

Coefficient reconstruction error: 3.44e-14.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Intersect the direct-ghost perpendicular bisector with each receive ray to estimate two surface contact points. Convert local reflector normals into slopes, sigma = -n_x/n_y. Two points with two slopes provide Hermite constraints for a cubic barrier curve, solved non-iteratively.

Implementation checkpoints

- Reject nearly coincident x1 and x2 because the solve becomes ill-conditioned.
- Scale x coordinates before inversion.
- Use pivoted linear solve rather than explicit matrix inverse.
- Bound curvature and extrapolation distance outside the two contact points.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A cloverleaf off-ramp rail curves from y = 3.0 m at x = 10 m to y = 5.2 m at x = 30 m. A planar model spreads ghost residuals along 132 m of rail. Two validated ghost pairs near x1 = 12 m and x2 = 24 m supply position and tangent constraints, reconstructing the local cubic to about 1e-14 numerical precision in under 0.1 ms.

6. THEORETICAL FOUNDATION - PHD LEVEL

Hermite interpolation of a C2 reflector using point and tangent constraints; confluent Vandermonde nonsingularity gives a closed local surface estimate.

Technical caution: A cubic is local, not a globally valid road model. Restrict its use to the supported interval and uncertainty envelope.

Distribution-Free Conformal Gating

NOVELTY SCORE 67.0 / 100 f360_determine_association_hypothesis.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

Instead of assuming every class follows a bell curve, set the passing threshold from the scores actually observed.

2. BASELINE FORMULATION

Gaussian chi-square gating

```
Assume d_M^2 follows chi-square_d
```

Heavy-tailed maneuver coverage: 0.042; RMSE: 17.683 m.

3. NOVEL FORMULATION

Non-conformity quantile gate

```
G_(1-alpha) = {z : S(z) <= q_hat_(1-alpha)}
S_i = sqrt(y_i^T S_i^-1 y_i)
q_hat from empirical (1-alpha)(1+1/n) quantile
```

Empirical coverage: 99.6%; RMSE: 0.263 m.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Maintain a calibration history of normalized innovation scores. Set the gate from the observed score quantile instead of a Gaussian theoretical quantile. This targets marginal empirical coverage even when residuals are skewed, multimodal, or heavy-tailed.

Implementation checkpoints

- Use an exchangeable or approximately stationary calibration window.
- Separate operating regimes if residual distributions differ strongly.
- Define tie handling and finite-sample quantile indexing explicitly.
- Monitor realized coverage and gate width online.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

During an S-curve slalom, a classical 9.21 threshold rejects 95 of 100 valid returns during transitions. The empirical residual distribution yields q_hat = 14.8 and retains 99.6% of valid detections.

6. THEORETICAL FOUNDATION - PHD LEVEL

Finite-sample conformal prediction using non-conformity scores and empirical quantiles, replacing parametric chi-square coverage assumptions.

Technical caution: Coverage guarantees weaken under severe temporal dependence or distribution shift. Combine with regime detection and bounded gates.

REPOSITORY 1 | RANK 6 | CONCEPT 6/25

Spectral Entropy Covariance Scaling

NOVELTY SCORE 67.0 / 100 Msmt_Update_ObjTrks_CTCA.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

In heavy spray, trust the vehicle motion history more than every scattered water echo, just as a driver holds a steady course through a brief splash.

2. BASELINE FORMULATION

Static Gaussian measurement covariance

```
R_k = R_0 = diag(sigma_r^2, sigma_theta^2, sigma_v^2)
```

Wet-weather tracking RMSE: 0.911 m.

3. NOVEL FORMULATION

Entropy-modulated Joseph covariance

```
R(t) = R_0 exp(beta H(t)/SNR(t))
H(t) = - sum_i p_i ln(p_i)
```

Tracking RMSE: 0.315 m, a 65.4% reduction.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Compute normalized Shannon entropy over Doppler-spectrum bins. Specular targets concentrate energy, whereas tire spray and diffuse surfaces distribute energy across many bins. As entropy rises relative to SNR, inflate measurement covariance. The Kalman gain falls and the tracker relies more strongly on its motion model.

Implementation checkpoints

- Normalize spectral power into probabilities with a numerical floor.
- Convert SNR to a consistent linear or dB domain before applying the model.
- Cap covariance inflation and preserve minimum observability.
- Use Joseph form for covariance stability.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

Behind a semi-truck in rain, returns span about +/-4 m/s with H = 2.85, SNR = 6 dB (linear 3.98), and beta = 4.0. The scale exp(4 x 2.85 / 3.98) is approximately 17.5. The reduced gain prevents heading oscillation driven by spray detections.

6. THEORETICAL FOUNDATION - PHD LEVEL

Information-theoretic measurement regularization; entropy lowers effective Fisher information and shifts the Bayesian update toward prediction.

Technical caution: Low SNR alone can make the ratio unstable. Apply SNR floors and entropy confidence checks.

Contamination-Budgeted Frozen Rank Gate

NOVELTY SCORE 63.0 / 100 f360_determine_association_hypothesis.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

A few students shouting nonsense should not change the classroom rules. Reserve room for known outliers and keep the boundary fixed.

2. BASELINE FORMULATION

Continuously sliding conformal threshold

```
Online quantile updates continuously.
```

Glint bursts inflate gate width from 1.58 to 30.0, admitting clutter.

3. NOVEL FORMULATION

Finite-budget rank sandwich

```
S_(231) <= q <= S_(247)
N = 256, replacement budget m = 8
Guaranteed lower coverage >= 0.9023
```

Gate remains bounded despite arbitrary bursts within the contamination budget.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Freeze calibration on an independent reference block. Allow up to m adversarially replaced scores and bound the admissible threshold between protected order statistics. Add a 2-of-3 sensor quorum before accepting gate adaptation.

Implementation checkpoints

- Select a clean reference interval and version it with the calibration.
- Choose the replacement budget from measured burst statistics.
- Do not update the frozen window during an active contamination alarm.
- Use quorum logic to avoid a single-sensor poisoning event.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A bridge expansion joint creates six glint spikes with d_M^2 above 100. A sliding gate widens to 30.0. The frozen rank gate reserves the upper eight positions and anchors the threshold near rank 247, keeping the effective boundary at or below 2.14 in the stated score scale.

6. THEORETICAL FOUNDATION - PHD LEVEL

Robust order statistics under Huber-style epsilon contamination; a finite replacement budget bounds quantile influence and gate inflation.

Technical caution: A stale reference block may under-cover a real regime change. Include controlled re-calibration criteria.

REPOSITORY 1 | RANK 8 | CONCEPT 8/25

Doppler-Invariant Ghost Disambiguation

NOVELTY SCORE 63.0 / 100 f360_detect_multipath.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

A statue may appear where a moving reflection should be, but its speed gives it away immediately.

2. BASELINE FORMULATION

Position-only geometric association

Associate candidate reflection using distance and geometry only.

Decoy precision: 0.638; false positives: 36.2%.

3. NOVEL FORMULATION

Radial Doppler projection gate

$$|v_{r,meas} - u_{spec}^T v| \le 3\sqrt{\sigma_v^2 + S_v}$$

Require confirmation in two scans.

Decoy precision: 0.877; ghost-rejection rate: 66.1%.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Project the tracked target velocity onto the predicted specular line of sight. Compare the predicted radial speed with each candidate return. Stationary roadside objects may match reflected position geometry but generally fail the Doppler consistency test.

Implementation checkpoints

- Propagate track velocity covariance into S_v.
- Use consistent radial-velocity signs and ego-motion compensation.
- Require two-scan confirmation to suppress isolated Doppler outliers.
- Relax the test at near-zero relative Doppler where discrimination is weak.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A vehicle moves at 25 m/s along a wall. The direct radar measures -24.8 m/s. A road sign at a similar mirrored distance measures -18.2 m/s, while the specular model predicts -20.19 m/s. The 1.987 m/s mismatch is reported as roughly 40 sigma under the stated noise model, so the sign is rejected.

6. THEORETICAL FOUNDATION - PHD LEVEL

Kinematic invariant formed by projecting target velocity onto the reflected line of sight; covariance propagation defines a statistically valid residual gate.

Technical caution: Velocity bias or poor ego-motion compensation can reject real ghosts. Calibrate Doppler bias before enabling a tight gate.

REPOSITORY 1 | RANK 9 | CONCEPT 9/25

IMM Acceleration Gating (Refuted Baseline)

NOVELTY SCORE 55.0 / 100 f360_determine_association_hypothesis.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

The key needed to open the box is locked inside the box: the mode cannot change before association, but association needs the mode to change first.

2. BASELINE FORMULATION

IMM probability-conditioned gate

`Use CV/CA model probabilities mu_j to scale the association gate.`

Observed recall: 88.0%, below the kinematics-aware result.

3. NOVEL FORMULATION

Retain direct kinematics-aware gating

`Acceleration must widen the gate before association.`

`IMM mode probability changes only after the point is associated.`

Refutation establishes a causal onset limitation; production recall remains approximately 99.9% with direct kinematic adaptation.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

The experiment demonstrated a circular dependency: maneuver-mode probability cannot increase until the new detection updates the IMM, but the detection cannot update the IMM unless the current gate accepts it. The result is documented as a negative research outcome rather than a production proposal.

Implementation checkpoints

- Retain the experiment as a regression test and design note.
- Do not use posterior IMM probabilities as the only onset signal.
- Use causal pre-association cues such as measured acceleration and innovation energy.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A vehicle suddenly cuts in. The CA mode should become likely, but the first maneuver detection lies outside the small CV gate. Because the detection is rejected, the mode probability never receives the evidence required to widen the gate.

6. THEORETICAL FOUNDATION - PHD LEVEL

Causal circularity in jump-Markov validation: posterior mode beliefs have onset latency and cannot safely control the pre-update acceptance region for abrupt transitions.

Technical caution: Reintroducing posterior-only IMM gating recreates the key-inside-the-box failure at maneuver onset.

Dual-Horizon Fast Dynamic Alignment

NOVELTY SCORE 70.0 / 100 dyn_align_wrapper.c

1. INTUITION - 13-YEAR-OLD LEVEL

Straighten crooked glasses quickly at first, then make only tiny corrections after the view becomes stable.

2. BASELINE FORMULATION

Static gain schedule

```
K(N) = K_0 approximately 0.0025
Startup deadband: N_wait = 400 frames
```

Lock may require 20 s at 20 Hz, and substantially longer when road-curve pauses are applied.

3. NOVEL FORMULATION

Dual-horizon exponential decay

```
K_DA(N) = K_fast exp(-alpha N) + K_slow
K_fast = 0.05, alpha = 0.08, K_slow = 0.0025
```

Convergence in at most 46 frames, below 2.3 s at 20 Hz.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Use a high initial correction gain and exponentially decay toward the stable production gain. Apply a stationary-clutter Doppler RANSAC gate, |v_r + v_ego cos(theta)| <= epsilon, so turns and moving objects do not corrupt boresight estimation.

Implementation checkpoints

- Cap startup updates in Auto_align_Init().
- Gate candidates with ego-compensated stationary Doppler.
- Switch or decay gain based on both frame count and residual stability.
- Persist converged alignment with validity metadata.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A vehicle exits a garage with azimuth error -1.145 degrees. The baseline requires 400 accepted frames; road-curvature pauses can extend elapsed time. The dual-horizon schedule makes large changes in frames 1-30 and locks when residual variance stabilizes at frame 46.

6. THEORETICAL FOUNDATION - PHD LEVEL

Robbins-Monro stochastic approximation with exponentially annealed step size, constrained by a robust zero-Doppler stationary-clutter manifold.

Technical caution: Aggressive startup gain can chase moving clutter. Require spatial diversity and robust inlier selection.

Track-Conditioned Residual Beat Estimator

NOVELTY SCORE 65.0 / 100 residual_track_ifc.c

1. INTUITION - 13-YEAR-OLD LEVEL

Tune directly near the expected radio station and adjust the fine knob instead of scanning the whole band.

2. BASELINE FORMULATION

Full 512-point range FFT

```
X[k] = sum_{n=0}^{N-1} x[n] W_N^(nk)
```

4,608 DSP multiplications per dwell; floating-point range RMSE: 0.439 m.

3. NOVEL FORMULATION

Lag-1 autocorrelation residual

```
f_res = angle(sum_{n=1}^{N-1} z[n]
conjugate(z[n-1])))/(2 pi T_s)
```

1,024 multiplications, 4.5x compute reduction; INT8 RMSE: 0.108 m.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Derotate raw fast-time ADC samples using the track-predicted center frequency. Estimate only the small residual frequency from the average lag-1 phase, enabling sub-bin refinement with low arithmetic cost and bounded INT8 implementation.

Implementation checkpoints

- Gate use by track confidence and predicted-bin error.
- Accumulate products with sufficient headroom.
- Handle phase wrapping and low-coherence cases.
- Retain a full FFT fallback for track acquisition.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

For a vehicle at 42.3 m with predicted beat frequency 1.41 MHz, the baseline runs a 512-point FFT plus interpolation. The residual method derotates the samples and evaluates lag-1 correlation in 1,024 operations, reaching 0.108 m RMSE under INT8 arithmetic in the stated test.

6. THEORETICAL FOUNDATION - PHD LEVEL

Model-predictive heterodyning followed by local phase-gradient maximum-likelihood frequency estimation on a complex analytic signal.

Technical caution: A bad predicted frequency can move the residual outside its unambiguous interval. Detect and revert to coarse FFT.

REPOSITORY 2 | RANK 3 | CONCEPT 12/25

Occupancy-BLUE CFAR with Pfa Control

NOVELTY SCORE 62.0 / 100 appl_cfar_ifc.c / appl_adaptive_cfar.c

1. INTUITION - 13-YEAR-OLD LEVEL

Listen less to uncertain witnesses, but do not throw away their information completely.

2. BASELINE FORMULATION

Uniform cell weighting

```
beta_i = 1/N
```

Reference cells are treated equally or removed by binary censoring, reducing sensitivity when partially occupied.

3. NOVEL FORMULATION

Variance-optimal BLUE weighting

```
beta_i proportional to m_i / v_i
Pfa(alpha) = product_i (1 + alpha beta_i)^-1
```

Target Pfa 1e-4 gives 1.02e-4; weak-target recall improves by 7.4%.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Use occupancy probability and expected cell statistics to soft-weight each reference cell. Solve the closed-form product equation for alpha by bounded bisection, retaining constant false-alarm behavior without large heuristic lookup tables.

Implementation checkpoints

- Normalize weights and bound extreme ratios.
- Solve alpha monotonically with fixed iteration limits suitable for firmware.
- Validate fixed-point quantization error in Pfa.
- Fallback to conventional CFAR if occupancy metadata is unavailable.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A 16-cell reference window overlaps two adjacent-vehicle cells with occupancy probability 0.85. Uniform averaging lets leaked target power inflate the noise estimate. BLUE assigns those cells a much smaller effective weight, stated as 0.08 relative to 1.0 for clean cells, improving weak pedestrian recall by 7.4%.

6. THEORETICAL FOUNDATION - PHD LEVEL

Minimum-variance unbiased estimation for heterogeneous exponential reference cells plus analytic Neyman-Pearson false-alarm calibration.

Technical caution: Miscalibrated occupancy probabilities can bias the threshold. Track Pfa by scene class and recalibrate weights.

REPOSITORY 2 | RANK 4 | CONCEPT 13/25

Range-Adaptive Dual-Loop Clutter CFAR

NOVELTY SCORE 62.0 / 100 appl_cfar_ifc.c

1. INTUITION - 13-YEAR-OLD LEVEL

Shield the glare from a flashlight so a nearby candle can still be detected.

2. BASELINE FORMULATION

Static-reference-window CA-CFAR

```
T(r) = alpha_CA / R_0 * sum_{i in W} P(r+i)
R_0 = 16
```

Edge recall: 50.0%; false alarms: 1 in the stated test.

3. NOVEL FORMULATION

Dual-loop background smoothing

```
C_hat_(k+1)(r) = (1-lambda) C_hat_k(r) + lambda median_d P(r,d)
s(r) = 0.5 |C_hat(r+1) - C_hat(r-1)|
R_0 in {8,16,32}, with clean-side censoring
```

Recall: 100.0%; false alarms: 0 in the stated test.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Maintain a slow background clutter floor using track-free Doppler bins. Detect abrupt range gradients at guardrail or wall edges, shrink the reference window, and censor cells on the contaminated side so the threshold is estimated from clean background.

Implementation checkpoints

- Update the background using only bins without active tracks or strong detections.
- Use hysteresis when changing window size.
- Maintain sufficient clean cells for stable Pfa control.
- Expose edge-state diagnostics for validation.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

At 1.2 m from a metal guardrail, a +45 dB return spills across six range bins. Static CA-CFAR raises the threshold by 18 dB and misses the nearby car. The adaptive method detects the slope, chooses R_0 = 8, excludes wall-facing cells, and recovers the target.

6. THEORETICAL FOUNDATION - PHD LEVEL

Multiscale spatiotemporal background estimation combining slow IIR statistics with gradient-guided unilateral censoring at clutter discontinuities.

Technical caution: Over-censoring reduces the statistical sample size and may destabilize Pfa. Enforce minimum reference-cell counts.

REPOSITORY 2 | RANK 5 | CONCEPT 14/25

Closed-Form Specular Contact Inversion

NOVELTY SCORE 60.0 / 100 ghost_calib.c

1. INTUITION - 13-YEAR-OLD LEVEL

A pool-ball bounce can be found from geometry directly instead of guessing the contact point repeatedly.

2. BASELINE FORMULATION

Iterative ray tracing

Numerically solve a nonlinear Fermat-path objective.

Iterative searches can exceed a 50 ms real-time cycle.

3. NOVEL FORMULATION

Closed-form algebraic solution

```
t = 2(l - u^T d) / (l^2 - ||d||^2)
Curvature certificate: ||g'|| <= 2 kappa_max rho_max
```

Contact-point RMSE: 0.018 m.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Solve the physical bounce point directly along receive unit vector u using path-length geometry. Evaluate a curvature limit that certifies whether the local planar-reflector assumption is valid; reject rather than force a solution outside that bound.

Implementation checkpoints

- Check denominator magnitude and physical range of t.
- Verify path-length closure after solving.
- Evaluate the curvature certificate before accepting calibration.
- Use double precision offline to generate fixed-point error bounds.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

For target p = (20, 2) m, sensor at the origin, wall distance 3.5 m, and total path length 28.42 m, the iterative baseline needs about 15 Newton steps. The closed form gives t = 8.12 m with six stated floating-point operations and 0.018 m error.

6. THEORETICAL FOUNDATION - PHD LEVEL

Algebraic inversion of a specular two-point boundary-value problem, with a curvature certificate delimiting the planar approximation.

Technical caution: Degenerate geometries can make the denominator small. Detect ill-conditioning and fall back safely.

REPOSITORY 2 | RANK 6 | CONCEPT 15/25

Priority-Ranked CDC Urban Packing

NOVELTY SCORE 60.0 / 100 cdc_packing.c / cdc_priority.c

1. INTUITION - 13-YEAR-OLD LEVEL

When a safety bag is full, pack medicine before decorative books.

2. BASELINE FORMULATION

First-come, first-served truncation

`Sequentially fill CDC output until MAX_CDC = 5016.`

Under saturation, late detections are dropped regardless of safety relevance.

3. NOVEL FORMULATION

Threat-weighted top-k selection

`score = SNR_linear / r^2`
`In-place O(N*k) selection for bounded firmware memory.`

Near-field and high-risk detections are packed before low-priority background.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Score range-Doppler candidates before Ethernet packaging. Reorder p_cdc_array and p_ci_flag_array with the same permutation, then serialize the highest-ranked detections into each 1472-byte UDP payload first.

Implementation checkpoints

- Use stable tie-breaking to keep replay deterministic.
- Apply floors to range and caps to SNR.
- Keep metadata arrays synchronized during swaps.
- Measure worst-case CPU time at maximum N and k.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A crowded intersection yields 6,200 detections, exceeding the 5,016 limit. Baseline truncation misses a pedestrian at 4 m. Its score is 31.62/16 = 1.98 for 15 dB SNR, while an 85 m building at 30 dB scores 1000/7225 = 0.138. The pedestrian is retained.

6. THEORETICAL FOUNDATION - PHD LEVEL

Capacity-constrained top-k selection over a hazard proxy proportional to SNR/r^2, preserving high-value detections under channel saturation.

Technical caution: The simple score may over-prioritize benign near-field clutter. Add class, closing speed, and confidence when available.

3-Product Cubic-Exact Residual Stencil

NOVELTY SCORE 57.0 / 100 `residual_track_ifc.c`

1. INTUITION - 13-YEAR-OLD LEVEL

A few carefully spaced wave samples can cancel the bend in frequency without fitting the entire curve.

2. BASELINE FORMULATION

Polynomial phase fitting

```
Fit phase coefficients by least squares across
many ADC samples.
```

Accurate but requires matrix operations and additional samples.

3. NOVEL FORMULATION

Three-complex-product stencil

```
eta_0 = angle(z_2 conjugate(z_1))
eta_+ = angle(z_3 conjugate(z_2))
eta_- = angle(z_1 conjugate(z_0))
f_hat = (6 eta_0 - eta_+ - eta_-)/(24 h)
```

Range RMSE: 13.13 mm with three complex products.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Use four lag-separated samples and three adjacent complex products. A fixed linear combination cancels modeled quadratic and cubic phase curvature, leaving the carrier-frequency term while avoiding least-squares inversion.

Implementation checkpoints

- Derive coefficients against the exact phase convention used in firmware.
- Select lag h to balance noise and phase wrapping.
- Use circular phase handling.
- Validate fixed-point coefficient quantization.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A vehicle accelerating at 8 m/s^2 adds chirp curvature and spreads a conventional estimator over about two frequency bins. The stencil cancels the modeled higher-order components and estimates the center frequency within 13.13 mm of range truth in the stated test.

6. THEORETICAL FOUNDATION - PHD LEVEL

Compact finite-difference annihilator for polynomial-phase components, designed for minimum complex-multiplication count.

Technical caution: The exactness depends on the assumed polynomial order and sample timing. Jerk or timing jitter can introduce bias.

REPOSITORY 3 | RANK 1 | CONCEPT 17/25

Preflight DA Header Injection

NOVELTY SCORE 70.0 / 100 sil_wrapper.cpp / Tracker_VariantA_Wrapper.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

Loading a saved game should restore completed setup instead of replaying the tutorial at every cluster slice.

2. BASELINE FORMULATION

Uninitialized cold start

```
theta_init = 0.0 degrees
```

Slice boundaries can trigger repeated convergence delays; the supplied estimate states 360,000 frames lost across 20 HPCC nodes.

3. NOVEL FORMULATION

Header-stream angle seeding

```
theta_0 = theta_header at t = 0
Read _DYNAMIC_ALIGNMENT_STREAM into
M2D_One_Time_Msg_T.
```

Startup latency drops to zero frames when a valid header value exists.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Read converged azimuth and elevation alignment from the log header and inject it into the radar configuration structure before frame 1. Each distributed replay slice begins from the recorded calibration state rather than replaying the convergence transient.

Implementation checkpoints

- Validate stream presence, version, endianness, and validity flags.
- Inject before any measurement-processing callback.
- Record the seeded value in replay metadata.
- Fallback explicitly to cold-start behavior when the header is absent.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

With a 300-log drive split over 20 nodes, Node 2 starts at Log 16. Baseline initialization uses 0.0 degrees and needs 400 frames, failing early UDP angle KPIs. Header injection seeds -1.145 degrees immediately.

6. THEORETICAL FOUNDATION - PHD LEVEL

Boundary-condition seeding for partitioned Markov trajectories; restores sufficient calibration state at each temporal subdomain boundary.

Technical caution: A stale or incompatible header can hide real alignment behavior. Include source timestamp and calibration version checks.

REPOSITORY 3 | RANK 2 | CONCEPT 18/25

Asynchronous Satellite Yaw Extrapolation

NOVELTY SCORE 68.0 / 100 sil_wrapper.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

Photos taken at slightly different times look doubled. Move every photo to the same timestamp and the objects line up.

2. BASELINE FORMULATION

Uncompensated timestamp stacking

```
p_fused = p_meas
Delta t in [5,45] ms
```

Coordinate error: 0.627 m to 0.711 m.

3. NOVEL FORMULATION

Kinematic rigid-body deskewing

```
p_corr = R_z(omega Delta t) p_meas - v_ego Delta t
- 0.5 a_ego Delta t^2
```

Fusion error reduction: 86.2% at omega = 0.1 rad/s and 31.7% at omega = 0.6 rad/s.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Transform each satellite detection from its measurement epoch to the central Domain Controller epoch. Apply yaw rotation plus translational ego-motion compensation to eliminate duplicated or smeared targets during turns.

Implementation checkpoints

- Define a single reference timestamp and frame convention.
- Interpolate ego yaw rate, velocity, and acceleration to each measurement time.
- Propagate timing and ego-motion uncertainty into detection covariance.
- Unit-test positive and negative Delta t.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

At 45 km/h in a roundabout with yaw rate 0.4 rad/s and 35 ms sensor skew, front-left and front-right detections of one vehicle differ by 0.68 m. Deskewing aligns them to about 0.09 m error.

6. THEORETICAL FOUNDATION - PHD LEVEL

Lie-group temporal deskew on SE(2), pulling asynchronous observations to a common epoch using the ego twist and translational kinematics.

Technical caution: High yaw-rate motion violates a constant-rate approximation over long delays. Use higher-order integration or shorter buffering when necessary.

Yaw-Free Timestamp and Doppler-Bias Separation

NOVELTY SCORE 60.0 / 100 sil_wrapper.cpp

1. INTUITION - 13-YEAR-OLD LEVEL

Use two moving references to set a slow watch without first fixing a tilted compass.

2. BASELINE FORMULATION

Coupled nonlinear estimation

`Jointly estimate time offset tau, Doppler bias b, and mounting yaw beta.`

The parameters can become confounded during acceleration and weak excitation.

3. NOVEL FORMULATION

Independent linear Gram invariant

`r_k w_k - d_k^T v_k = ||v_k||^2 tau`

Separates tau and b to about 1e-12 numerical precision; timing RMSE: 3.8 ms.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Combine measured range and range rate with reference trajectory vectors so the unknown mounting-yaw rotation cancels from the timing equation. Solve a two-target Gram system for clock offset and Doppler bias, then estimate yaw after temporal correction.

Implementation checkpoints

- Require targets with sufficiently independent velocity vectors.
- Check the Gram matrix condition number.
- Estimate Doppler bias jointly only after units and signs are normalized.
- Solve yaw after timestamp correction.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A satellite radar has 15 ms clock delay and 0.8 degree yaw error. The nonlinear baseline trades one error against the other. The invariant equation removes yaw from the timing solve and recovers the 15 ms offset independently.

6. THEORETICAL FOUNDATION - PHD LEVEL

Gauge elimination of SO(2) mounting yaw through Gram inner products, exposing timing and Doppler parameters in a linear identifiable subsystem.

Technical caution: Poor target geometry makes the Gram system singular or noisy. Add excitation checks before calibration.

REPOSITORY 3 | RANK 4 | CONCEPT 20/25

Cross-Satellite Relative Yaw Cancellation

NOVELTY SCORE 58.0 / 100 `sil_wrapper.cpp`

1. INTUITION - 13-YEAR-OLD LEVEL

Two observers compare the same object and reflection; shared motion cancels and the angle between observers remains.

2. BASELINE FORMULATION

Clutter-dependent alignment

```
Estimate boresight from stationary roadside features.
```

Feature-sparse highways can leave drift unobserved by up to 13.0 degrees.

3. NOVEL FORMULATION

Direct-ghost difference phase

```
D_i,k = exp(-j beta_i) (p_k - p*_k)
delta_hat_ij = arg(sum_k D_i,k conjugate(D_j,k))
```

Relative boresight yaw recovered to 1e-10 rad in the stated synthetic test.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

For overlapping sensors, subtract direct and ghost vectors. This cancels common target motion and much of the wall geometry. Cross-correlating the complex difference vectors isolates the relative sensor rotation.

Implementation checkpoints

- Confirm that direct and ghost observations share the same parent.
- Accumulate across multiple frames and weight by covariance.
- Reject low-magnitude difference vectors.
- Anchor the relative network with one absolute calibration source.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

On an open desert highway without poles or signs, independent alignment drifts. Front-right and rear-right sensors observe a passing vehicle and its rail reflection. Cross-multiplication cancels the trajectory and estimates relative yaw without stationary point clutter.

6. THEORETICAL FOUNDATION - PHD LEVEL

Gauge-invariant differential phase correlation cancels common target translation and reflector orientation, isolating relative SO(2) sensor yaw.

Technical caution: Incorrect ghost pairing directly corrupts relative yaw. Use strict temporal and Doppler consistency.

REPOSITORY 4 | RANK 1 | CONCEPT 21/25

4D Doppler-RadarSplat Generative Simulation

NOVELTY SCORE 70.0 / 100 `models/doppler_radarsplat.py`

1. INTUITION - 13-YEAR-OLD LEVEL

A radar is not a blurry camera. It measures echo strength, distance, direction, and the pitch shift caused by motion.

2. BASELINE FORMULATION

Optical image synthesis

```
Render RGB or depth approximations; omit
Doppler, RCS, and CDC spectra.
```

Insufficient for firmware-level Range-Doppler validation.

3. NOVEL FORMULATION

Differentiable RF primitives

```
v_r,i = (v_i - v_ego)^T u_i
P_r proportional to sigma_RCS / R^4
```

Analytic gradient agreement below 1e-8; velocity RMSE: 0.036 m/s in the stated comparison.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Extend dynamic Gaussian splats with radar cross section, range-dependent attenuation, line-of-sight Doppler, and spectral spread. Rasterize them into synthetic Range-Doppler compressed data cubes suitable for firmware and tracker testing.

Implementation checkpoints

- Keep all RF operations differentiable for parameter fitting.
- Model antenna pattern and occlusion, not only R^-4 loss.
- Calibrate RCS distributions by class and aspect.
- Validate both aggregate KPIs and individual spectral signatures.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A car travels at 30 m/s at 40 m range. Optical simulators output shape and depth only. RadarSplat applies R^-4 power loss and radial projection to synthesize a CDC spectrum whose velocity matches logs within 0.036 m/s RMSE in the stated test.

6. THEORETICAL FOUNDATION - PHD LEVEL

Differentiable RF field synthesis on SE(3) x R; Gaussian primitives projected through radar-equation attenuation, RCS, antenna response, and Doppler operators.

Technical caution: A visually plausible synthetic scene can still be electromagnetically wrong. Validate against measured CDC distributions.

Associative Parallel Prefix-Scan Kalman Filter

NOVELTY SCORE 64.0 / 100 engine/parallel_kf.py

1. INTUITION - 13-YEAR-OLD LEVEL

Add pairs at the same time, then pairs of pairs. A long chain becomes a short tree.

2. BASELINE FORMULATION

Serial time-stepping loop

```
x_k = F_k x_(k-1) + K_k y_k
```

50,000 frames require 50,000 dependent steps and form a replay bottleneck.

3. NOVEL FORMULATION

Associative binary-tree scan

```
S_(i,j) = S_(i,k) tensor S_(k+1,j)
50,000 steps map to 16 parallel levels.
```

Span reduction: 3,125x; relative numerical error: 6.35e-15.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Represent each filtering interval as an associative information-form element (A, b, C, eta, J). Combine adjacent intervals using a Blelloch prefix scan so GPU workers process a binary tree instead of a serial chain.

Implementation checkpoints

- Prove and unit-test associativity for the exact element representation.
- Use stable matrix factorizations and symmetry enforcement.
- Support segmented scans for track births and deaths.
- Compare against a serial oracle across random seeds.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A 50,000-frame vehicle track takes 45 s in a serial replay. The stated GPU prefix version processes 16 tree levels in under 0.02 s while matching serial EKF position to 1e-10 m.

6. THEORETICAL FOUNDATION - PHD LEVEL

Associative algebra of Gaussian information elements and Blelloch scans reduces serial span from O(T) to O(log T).

Technical caution: Numerical associativity is approximate in floating point. Reduction order and ill-conditioned covariances can affect reproducibility.

REPOSITORY 4 | RANK 3 | CONCEPT 23/25

Ghost-Conditioned Generative Target Augmentation

NOVELTY SCORE 64.0 / 100 `models/ghost_augmentation.py`

1. INTUITION - 13-YEAR-OLD LEVEL

An AI trained only on clean scenes gets confused by mirrors. Realistic reflection examples teach it what ghosts look like.

2. BASELINE FORMULATION

Uniform synthetic insertion

`Insert reflections without dynamic-range balancing; stated power spread is about 500x.`

False-alarm rate: 0.425.

3. NOVEL FORMULATION

Physics-normalized ghost injection

`Generate specular geometry and normalize power with tracker-predicted range and reflector properties.`

False-alarm rate: 0.033; recall: 1.000.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Generate multipath geometry inside the Doppler-RadarSplat pipeline. Scale reflection energy using range, incidence angle, guardrail conductivity, and predicted target power so the synthetic distribution does not overwhelm the learner.

Implementation checkpoints

- Condition on scene geometry, reflector material, and target state.
- Match real-log amplitude and entropy distributions.
- Separate training augmentation from validation data generation.
- Track performance by range and ghost type.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

Targets at 5 m can be about 500 times stronger than targets at 70 m under the chosen scenario. Uniform insertions teach a detector to fire on synthetic artifacts, producing 42.5% false alarms. Physics normalization reduces false alarms to 3.3% while preserving 100% recall.

6. THEORETICAL FOUNDATION - PHD LEVEL

Physics-conditioned generative priors regularize training distributions against near-field dynamic-range and specular multipath artifacts.

Technical caution: Normalization can accidentally leak tracker labels into training. Keep causal feature boundaries explicit.

REPOSITORY 4 | RANK 4 | CONCEPT 24/25

Nonlinear Parallel Gram-Lifted State Space

NOVELTY SCORE 60.0 / 100 engine/gram_lift.py

1. INTUITION - 13-YEAR-OLD LEVEL

Introduce distance-squared as a new variable so a curved equation becomes a straight one that parallel workers can process.

2. BASELINE FORMULATION

Iterated extended Kalman smoothing

Repeatedly relinearize nonlinear measurements around a trajectory.

Outer iterations create sequential dependencies across sweeps.

3. NOVEL FORMULATION

Quadratic invariant state lifting

zeta = [||p||^2, p^T v, ||v||^2]^T
Use Pascal-matrix transitions for polynomial kinematics.

Linear measurement closure error: 5.87e-13.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Augment the state with quadratic inner-product invariants. Range-squared and range-rate relations become linear in the lifted state, allowing direct use within an associative prefix scan without outer relinearization loops.

Implementation checkpoints

- Define closure for every transition and process-noise term.
- Control lifted-state covariance growth.
- Project back to physically consistent p and v states.
- Compare lifted and iterated solutions on strongly nonlinear cases.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A maneuvering vehicle is observed through range, bearing, and radial speed. Conventional parallel smoothing still requires trajectory-dependent relinearization. The lifted invariants make range-squared updates linear and preserve parallel execution.

6. THEORETICAL FOUNDATION - PHD LEVEL

Koopman-inspired finite lifting with quadratic invariants and Pascal transitions, obtaining exact polynomial closure for range-squared dynamics.

Technical caution: State lifting increases dimension and may admit inconsistent invariant combinations. Use constraint projection or regularization.

REPOSITORY 4 | RANK 5 | CONCEPT 25/25

Guarded Error-Transfer Replay Certificates

NOVELTY SCORE 58.0 / 100 `engine/guarded_replay.py`

1. INTUITION - 13-YEAR-OLD LEVEL

Before jumping into the middle of a movie, prove that the missing beginning cannot change the current plot decision.

2. BASELINE FORMULATION

Empirical warm-up windows

`Prepend a guessed warm-up interval, for example 200 frames.`

State and track-ID divergence remains possible near slice boundaries.

3. NOVEL FORMULATION

Guarded scalar affine operators

`Represent interval as S = (a, b, h).`
`A certified radius e < 1/3 preserves discrete branch equivalence.`

Enables slice equivalence checks without executing arbitrary warm-up frames.

4. TECHNICAL MECHANISM AND IMPLEMENTATION

Summarize an interval using affine error transfer and explicit distances to branch boundaries, such as track-birth or deletion thresholds. If propagated error remains inside the guard margin, the sliced replay is certified to make the same discrete decisions as continuous execution.

Implementation checkpoints

- Enumerate every discrete branch that affects externally visible state.
- Compute conservative error-transfer bounds.
- Fail closed when a guard cannot be certified.
- Store certificate results alongside replay KPIs.

5. CONCRETE SCENARIO AND NUMERICAL EXAMPLE

A log is split at frame 10,000. A track born at 9,950 may not be reconstructed by a guessed 200-frame warm-up if prior state differs. The certificate evaluates the decision guard, stated as $h_k > 0.33$, and proves whether the slice preserves the continuous birth/death sequence.

6. THEORETICAL FOUNDATION - PHD LEVEL

Formal verification of a hybrid dynamical system using affine error transfer and decision-margin guards across temporal domain decomposition.

Technical caution: A certificate is only as complete as the modeled branches. Unmodeled saturation, quantization, or nondeterminism can invalidate it.

Overall Novelty Index

Overall rank	Breakthrough	Repository	Score
1	**Virtual Aperture Specular Tracking**	1	80.0
2	**Rank-1 Specular Invariant and Bearing Correction**	1	74.0
3	**Kinematics-Aware Adaptive Gating**	1	71.0
4	**4D Doppler-RadarSplat Generative Simulation**	4	70.0
5	**Dual-Horizon Fast Dynamic Alignment**	2	70.0
6	**Preflight DA Header Injection**	3	70.0
7	**Asynchronous Satellite Yaw Extrapolation**	3	68.0
8	**Tangent-Mirror Hermite Linearization**	1	68.0
9	**Distribution-Free Conformal Gating**	1	67.0
10	**Spectral Entropy Covariance Scaling**	1	67.0
11	**Track-Conditioned Residual Beat Estimator**	2	65.0
12	**Associative Parallel Prefix-Scan Kalman Filter**	4	64.0
13	**Ghost-Conditioned Generative Target Augmentation**	4	64.0
14	**Contamination-Budgeted Frozen Rank Gate**	1	63.0
15	**Doppler-Invariant Ghost Disambiguation**	1	63.0
16	**Occupancy-BLUE CFAR with Pfa Control**	2	62.0
17	**Range-Adaptive Dual-Loop Clutter CFAR**	2	62.0

Overall rank	Breakthrough	Repository	Score
18	**Closed-Form Specular Contact Inversion**	2	60.0
19	**Nonlinear Parallel Gram-Lifted State Space**	4	60.0
20	**Priority-Ranked CDC Urban Packing**	2	60.0
21	**Yaw-Free Timestamp and Doppler-Bias Separation**	3	60.0
22	**Cross-Satellite Relative Yaw Cancellation**	3	58.0
23	**Guarded Error-Transfer Replay Certificates**	4	58.0
24	**3-Product Cubic-Exact Residual Stencil**	2	57.0
25	**IMM Acceleration Gating (Refuted Baseline)**	1	55.0